

K-Means Clustering

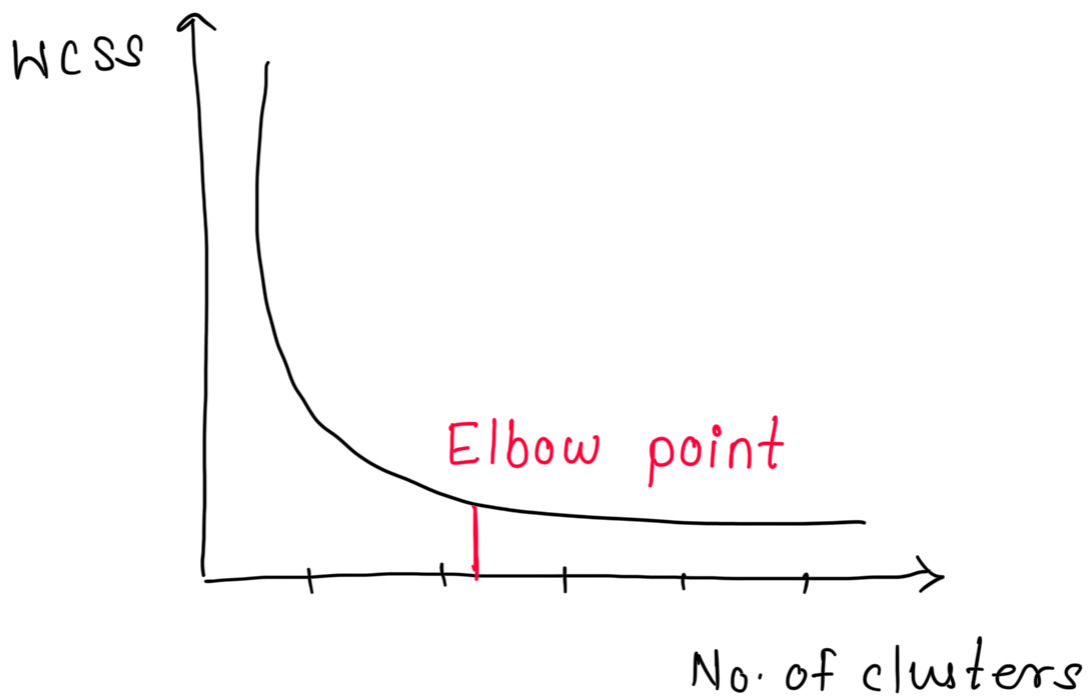
It is a popular unsupervised ML algorithm that partitions data into K distinct, non-overlapping subgroups (clusters) based on feature similarity.

It iteratively assigns data points to the nearest centroid and updates centroids to the mean of assigned points, minimizing within cluster variance

Algorithm steps

- 1) Decide n cluster
- 2) Init centroids
- 3) Assign cluster
- 4) Move centroids
- 5) Finish

Selecting k



The optimal number of cluster (k) is often found using the Elbow Method which plots the total within-cluster sum of squares (variance) against k and looks for the elbow where the rate of decrease drops. Another metric is Silhouette Coefficient.